OOS Number: OOS- 262335

Phase I

Section A – Test Details					
Initiator:	Guanchen Li (Written by: Dhvanir Kansara)	**Test Date:**	05-Oct-2026	**Date Initiated:**	13-Oct-2026
Date of Incident:	13-Oct-2026	**Test Name:**	USP <71> / EP 2.6.1 Sterility Test	**Sample ID:**	ETX-260902-0364 (BUD-T30)
Name of Analyst who Performed the Test:	Prepping Analyst: Mukyung Jang (MJ) Processing Analyst: Guanchen Li (GL) Reading Analyst: Elysse Nioupin (EN)	**Sample / Active Name**	TetraVisc™ Plus (TVP-002)	**Lot #:**	2026082002
		Dosage Form	Liquid		
Description of Incident:	On 14 Oct 2026, sample ETX-260902-0364 (BUD-T30) was found positive for viable microorganisms after USP <71> sterility testing.	☑ Client Care notified of OOS ☑ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method #: MICRO-SOP-5	Effective Date: 24Jul2026	SOP / Test Method Rev: 19	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 13-Oct-2026	Quality Assurance Department notified:	Yes ☑ No ☐	

Section B – Investigation and Root Cause Analysis			
Personnel Discrepancy		**Comments**	
Analyst interviewed?	☑ Yes ☐ No ☐ N/A	Yes, analysts Mukyung Jang, Guanchen Li, Elysse Nioupin were interviewed comprehensively.	
Correct samples analyzed?	☑ Yes ☐ No ☐ N/A	Yes, sample ID: ETX-260902-0364 (BUD-T30)	
Correct SOP / Test Method used/followed?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Correct sample collection/technique?	☑ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Are the Test Result Records Attached & Calculations Verified and Correct?	☑ Yes ☐ No ☐ N/A	Yes, sample ID: ETX-260902-0364 (BUD-T30)	
Is the analyst qualified to perform the test?	☑ Yes ☐ No ☐ N/A	Yes, the analysts are trained and qualified by quality to perform the test	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☑ N/A	Not Applicable	
Was the sample stored appropriately?	☑ Yes ☐ No ☐ N/A	Yes, the sample was stored as per client's instructions. Information is available in EagleTrax Sample Location History under ETX-260902-0364 (BUD-T30)	
Material Discrepancy		**Lot Number**	**Expiry Date**
Is the integrity of the sample(s) questionable?	☐ Yes ☑ No ☐ N/A		
Was the sample transported appropriately?	☑ Yes ☐ No ☐ N/A		
Are the other test results from this run under investigation?	☐ Yes ☑ No ☐ N/A		

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Appropriate/correct glassware used?	☐Yes ☐ No ☑N/A		
Are reagents stored in proper condition and not expired?	☑Yes ☐ No ☐N/A	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	Not applicable	Not applicable	Not applicable
Solvent Used	Not applicable	Not applicable	Not applicable
Is the standard the source of the root cause?	☐Yes ☐ No ☑N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐Yes ☐ No ☑N/A		
Equipment Discrepancy		**Equipment ID**	**Calibration Due Date**
Was the instrument set up correctly?	☑Yes ☐ No ☐ N/A	ISO 5 BSC E001314	Jun 2027
Is the clean room facility certified?	☑Yes ☐ No ☐ N/A	E001737 (CR115)	Dec 2026
System Performance Check (if applicable)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 2% (PDA)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
System Suitability RSD < 10% (QDA, CAD)?	☐Yes ☐ No ☑ N/A	Not applicable	Not applicable
Do the environmental monitoring results conform to required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☑Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☑Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor E001449) Incubator E001034 (Monitored by Sensor E001501) Incubator E001031 (Monitored by Sensor E001505) ISO 5 BSC E001314	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027 Feb 2027 Jun 2027

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Are chromatogram/spectra/raw data correct?	☐ Yes ☐ No ☑ N/A	Not applicable	Not applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

- ☑ N/A
- ☐ Re-injection of sample vial
- ☐ Re-injection of original standard vial
- ☐ Re-vialing of original working standard solution
- ☐ Re-vialing of original working sample solution
- ☐ Re-dilution of a working sample solution from the original stock
- ☐ Re-dilution of a working standard solution from the original stock standard
- ☐ Other: N/A DK 28Oct26

Section D – Phase I Summary

Phase I Summary:

All analysts involved in the prepping, processing, and reading of the sample – Mukyung Jang, Guanchen Li, Elysse Nioupin were interviewed comprehensively. Their answers are recorded throughout this document. This investigation was performed as per MICRO-SOP-53 Sterility Test Out-of-Specification (OOS) Investigation Procedure.

Upon arrival, samples were stored in accordance with the Client's instructions. Analysts - Mukyung Jang and Guanchen Li - verified the integrity of the samples throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of samples.

All reagents and supplies mentioned in the material section above were stored according to the suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system.

The cleanroom suite (CR115) used for processing procedures comprises three interconnected sections: the innermost ISO 7 cleanroom (115B), which connects to the middle ISO 7 buffer room (115A), and then to the outermost ISO 8 anteroom (115). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 115B through 115A and into 115.

The ISO 5 BSC E001314, located in the innermost ISO 7 room, (115B), was used for sample processing steps. It was thoroughly cleaned and disinfected prior to each procedure in accordance with MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Additionally, BSC E001314 was certified and approved by both the Engineering and Quality Assurance teams. Sample processing was conducted in the ISO 5 BSC E001314 in the innermost ISO 7 room, (115B) by Guanchen Li on 05 Oct 2026 as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

On 05 Oct 2026, the sample vials for ETX-260902-0364 (BUD-T30) were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001314.

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Phase I Summary (continued):

Inside the BSC, each container was aseptically opened, ensuring minimal exposure to the environment. The contents were then dispensed directly from the container into media bottles to perform Direct Inoculation. The sample was processed and tested as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test). Media bottles were transferred into incubators E001356 and E001357 to begin incubation.

On 13 Oct 2026 (Day 8 of incubation), microbial growth was observed in one TSB (Tryptic Soy Broth) media bottle by analyst, Elysse Nioupin, and confirmed by Microbiology Lab Supervisor, Robin Seymour. The positive TSB bottle for ETX-260902-0364 (BUD-T30) was submitted for Differential Staining and Microbial Identification under ETX-261013-0348. Differential staining indicated the presence of Gram-positive rods. Definitive identification identified the isolate as Bacillus licheniformis (Gram (+) rods).

Upon review of the environmental monitoring results, no microbial growth was detected on personnel monitoring plates collected on the date of testing (05 Oct 2026), the preceding testing date (03 Oct 2026), or the subsequent testing date (06 Oct 2026). No microbial growth was detected on settling plates or ISO 5 BSC surface sampling plates collected on the date of testing (05 Oct 2026), the preceding testing date (02 Oct 2026), or the subsequent testing date (06 Oct 2026). Additionally, no microbial growth was recovered from the weekly surface sampling of the anteroom, buffer room, or cleanroom locations collected on 29 Sep 2026.

Weekly active air sampling collected on 29 Sep 2026 recovered 9 CFUs consisting of Gram-positive cocci and fungal hyphae from the ISO 8 Anteroom (115) (ETX-261006-0664). During the week of testing, weekly active air sampling collected on 06 Oct 2026 recovered 1 CFU identified as Corynebacterium segmentosum (Gram-positive short rods) from ISO 7 Buffer Room 115A (ETX-261013-0297).

The Corynebacterium segmentosum recovered from the ISO 7 Buffer Room active air sample exhibited morphological characteristics consistent with the Gram-positive rods recovered from the test sample; however, it was not microbiologically identical to the organism associated with the observed test sample growth. Additionally, no microbial growth was recovered from personnel monitoring samples, ISO 5 BSC work surface samples, or passive air monitoring samples collected during the testing period. Furthermore, the biological safety cabinet used for sample processing is located within a separate, enclosed ISO 7 processing room (115B) internal to the buffer room suite, providing an additional physical and environmental segregation barrier between the active air sampling location and the ISO 5 processing environment.

The absence of contamination on work-surface environmental monitoring plates and the lack of contamination in other samples processed during the same testing session indicate that the ISO 5 testing environment remained in a state of control and do not support viable transfer of contamination from the surrounding cleanroom environment into the test system during sample processing.

Therefore, based on the absence of microbial recovery from ISO 5 critical-area monitoring and personnel monitoring, the lack of microbiological correlation between organisms recovered from environmental monitoring and the isolate recovered from the test sample, and the absence of growth on weekly surface monitoring plates from Cleanroom Suite CR115, the cleanroom environment is not considered a credible source of the Bacillus licheniformis recovered from the test sample.

The analysts confirmed full compliance with cleaning procedures as outlined in MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

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Phase I Summary (continued):

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 BSCs for Cleanroom suite CR115 was performed on 13 Sep 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). During both cleaning cycles, it was documented that all H_2O_2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom suite CR115.

To evaluate the potential for sample-to-sample cross-contamination, all samples processed on 05 Oct 2026 were reviewed. The processing analyst handled three additional stability samples from the same client during the same testing session, encompassing both room-temperature and accelerated storage conditions (Table 3). No other samples processed on the testing date exhibited microbial growth. Consequently,these findings provide no evidence of cross-contamination during sample processing and indicate that sample-to-sample contamination was unlikely to have contributed to the positive sterility result.

A review of the six-month testing history for Ocusoft (E12950) identified no previous sterility test failures under USP <71> / EP 2.6.1. The baseline T0 sample, tested on 03 Sep 2026 by a different analyst, met all sterility acceptance criteria. The same sample lot associated with the positive result (Lot 2026082002) was subsequently tested on 05 Oct 2026 as part of the BUD stability study by the processing analyst, Guanchen Li. This companion sample, which had been stored under accelerated conditions (40°C), was processed during the same testing session using the same equipment, materials, and aseptic practices and yielded a negative sterility result. The negative sterility result obtained from a same-lot sample processed under comparable laboratory conditions does not support analyst-related contamination, laboratory error, or a systematic processing deficiency as the source of the positive result. Furthermore, although microbial growth was observed in the room-temperature sample, no microorganisms were recovered from the companion same-lot sample tested during the same session.

Based on the observations outlined above, it is unlikely that the failing results were due to reagents, supplies, the cleanroom environment, the process, or analyst involvement. Consequently, the possibility of laboratory error contributing to this failure is minimal and the original result is deemed to be valid.

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Has root cause been identified as a possible laboratory error?
Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes

☐ If retesting is ***required***, complete Phase II of the OOS (initiate Form 3.100.019.F02).

☑ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (*Print*)**:** Dhvanir Kansara	Signature/Date:
Lab Manager/Designee (*Print*)**:**	Signature/Date:

Section E – Quality Disposition

Can investigation be closed?

☐ Yes

☐ Original result is valid.

☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (*Print*)**:**	Signature/Date:

Table 1: Information for samples under investigation OOS-262335

Reading Analyst	Day of Incubation	Sample ID	Related Microbial ID	Media with microbial growth	Microbial Identification
Elysse Nioupin	08	ETX-260902-0364 (BUD-T30)	ETX-261013-0348	1 x 100mL TSB	*Bacillus licheniformis* Gram (+) rods

Table 2: Environmental Monitoring for Analyst & Cleanroom

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMMYYYY)	Analyst (Initials)	Day /Week(s)	Observation	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing 05 Oct 2026								
Personnel (Left Touch and Right-Hand Touch)	Daily	03 Oct 2026	GL	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right- Hand Touch)	Daily	05 Oct 2026	GL	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right- Hand Touch)	Daily	06 Oct 2026	GL	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC) E001314								
Surface Sampling of ISO 5 (4 locations)	Daily	02 Oct 2026	AOD	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	05 Oct 2026	GL	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	06 Oct 2026	AOD	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	02 Oct 2026	AOD	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	05 Oct 2026	GL	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	06 Oct 2026	AOD	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling for CR 115 (E001737)								
Active Air Sampling of Cleanrooms	Weekly	29 Sep 2026	SMO	Week Before Testing	9 CFU from ISO 8 115	ETX-261006-0664	Gram (+) cocci Fungal hyphae	None
Active Air Sampling of Cleanrooms	Weekly	06 Oct 2026	SMO	Week of Testing	1 CFU from ISO 7 115A	ETX-261013-0297	*Corynebacterium segmentosum* Gram (+) short rods	None
Surface Sampling of Anteroom, Buffer room and Cleanroom for CR 115 (E001737)								
Surface Sampling of Cleanrooms	Weekly	29 Sep 2026	SMO	Week Before Testing	No growth	N/A	N/A	None
Surface Sampling of Cleanrooms	Weekly	06 Oct 2026	SMO	Week of Testing	No growth	N/A	N/A	None

Table 3: Client Samples Processed on the Same Day by Processing Analyst

ETX -Submission ID	Timepoint	Sample Name	Lot:	Storage Conditions	Sterility Reads
ETX-260902-0668	T30	TetraVisc™ Plus (TVP-011)	2026082001	Room Temperature	Negative
ETX-260902-0364	T30	TetraVisc™ Plus (TVP-002)	2026082002	Room Temperature	Positive for Microbial Growth
ETX-260902-0752	T30	TetraVisc™ Plus (TVP-011)	2026082001	Accelerated – 40 degree Celsius	Negative
ETX-260902-0582	T30	TetraVisc™ Plus (TVP-002)	2026082002	Accelerated – 40 degree Celsius	Negative